

B.TECH 3RD SEM DISCRETE MATHEMATICS

BEU PATNA

OBJECTIVE

QUESTION



LAKSHYA BATCH *FOR CSE*



Start Date:

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4TH SEMESTER



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1. If A has n elements, then the number of elements in the power set $P(A)$ is:

(a) n^2

(b) 2^n

(c) n^n

(d) $2n$

$$|P(A)| = 2^n$$

$$A = \{1, 2\}$$
$$|A| = 2 \quad \rightarrow \quad 2^2 = 4$$

$$P(A) = \{ \emptyset, \{1\}, \{2\}, \{1, 2\} \}$$

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2. A relation R on set A is an equivalence relation if it is:

- (a) Reflexive and Symmetric
- (b) Symmetric and Transitive

- ☒ (c) Reflexive, Symmetric, and Transitive
- (d) Reflexive and Transitive

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3. If $f : A \rightarrow B$ is both one-to-one (injective) and onto (surjective), then f is called:

- (a) Composite
- (b) Identity

- (c) Bijective
- (d) Constant

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4. The GCD of two prime numbers p and q (where $p \neq q$) is always:

- (a) $p \cdot q$
- (b) 0

- ☒ (c) 1
- (d) $p + q$

$$\begin{array}{l} 2 = 2 \times 1 \\ 3 = 3 \times 1 \end{array} \rightarrow \text{HCD} = 1$$

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5. According to the Pigeonhole Principle, if $n + 1$ pigeons are put into n holes, then at least one hole contains at least:

(a) 1 pigeon

☒ (b) 2 pigeons

(c) n pigeons

(d) 0 pigeons

$$\boxed{n+1} = 5$$

→ Pigeons

$$\frac{n}{4} = 4$$

Pigeon holes



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6. The number of ways to arrange n distinct objects in a circle is:

(a) $n!$

(b) $(n + 1)!$

~~(c)~~ $(n - 1)!$

(d) n^2



A circle with 10 tick marks representing objects. A yellow arrow points from one tick mark to the next, labeled 'n!'. To the right of the circle, the following calculation is written:

$$\frac{(n-1)! \cdot n}{n} = (n-1)!$$

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7. Which of the following is a Tautology?

(a) $p \wedge \neg p$

(b) $p \vee \neg p$

(c) $p \rightarrow \neg p$

(d) $p \wedge q$

Ⓐ $p \wedge \neg p$
 $T \wedge F = F$

Ⓑ $p \vee \neg p$
 $T \vee F = T$

p	$\neg p$
T	F

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8. The negation of the statement "All doctors are rich" is:

(a) All doctors are poor

(b) No doctor is rich

☒ (c) Some doctors are not rich

(d) Some doctors are rich

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9. In a simple graph with n vertices, the maximum number of edges is:

- (a) n
- (b) $n(n - 1)$

- ☒ (c) $n(n - 1)/2$
- (d) n^2

$$E = \frac{n(n-1)}{2}$$

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10. A graph in which all vertices have the same degree is called:

(a) Complete Graph

(b) Regular Graph

(c) Bipartite Graph

(d) Connected Graph

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11. The sum of degrees of all vertices in a graph is equal to:

(a) Number of edges

(c) Half the number of edges

☒ (b) Twice the number of edges

(d) Number of vertices

$$\sum \deg(v) = 2|E|$$

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12. A tree with n vertices has exactly how many edges?

(a) n

(b) $n + 1$

 (c) $n - 1$

(d) $2n$

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13. An algebraic structure $(G, *)$ is a group if it satisfies:

- ☒ (a) Closure, Associativity
- ☒ (b) Identity, Inverse

- ☒ (c) Both (a) and (b)
- ☐ (d) None of the above

- ✓ Closure, Associativity
- ✓ Identity, Inverse

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14. The smallest non-Abelian group has an order of:

(a) 3

(b) 4

(c) 5

☒ (d) 6

$1, \underline{2}, 3, 4, \underline{5}, S_3 = 3! = 3 \times 2$

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15. In a Boolean Algebra, $x + x$ is equal to:

(a) $2x$

(b) 0

(c) 1

☒ (d) x

$$1+1=1$$

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16. The identity element for the 'OR' operation in Boolean Algebra is:

(a) 0
(b) 1

(c) x
(d) x'

$$\begin{array}{l} \text{OR} \\ 1 + 0 = 1 \\ 0 + 1 = 1 \end{array}$$

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17. A path that visits every ^{vertices} edge exactly once is called an: ^{→ Hamiltonian}
- (a) Hamiltonian Path
 - ✓ (b) Euler Path
 - (c) Planar Path
 - (d) Bipartite Path

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18. The number of edges in a Complete Graph K_5 is:

(a) 5
(b) 10

(c) 20
(d) 25

$$|E| = \frac{n(n-1)}{2} = \frac{5(5-1)}{2} = \frac{20}{2} = 10$$

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19. Euler's formula for a planar graph is given by:

(a) $V - E + F = 1$

☒ (b) $V + E - F = 2$

(c) $V - E + F = 2$

(d) $V + E + F = 2$

$$V + E - F = 2$$

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20. Which of the following is true for a Cyclic Group?

☒ (a) It is always Abelian

☐ (b) It is never Abelian

☐ (c) It has no generator

☐ (d) It must be infinite

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21. The dual of the Boolean expression $x \cdot (y + 0)$ is: $= x + (y \cdot 1)$

☒ (a) $x + (y \cdot 1)$

☐ (b) $x' \cdot (y' + 1)$

☐ (c) $x + (y + 1)$

☐ (d) $x \cdot y$

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22. If the GCD of a and b is 1, then a and b are said to be:

(a) Even numbers

(b) Composite numbers

☒ (c) Relatively Prime

(d) Twin primes

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23. A complete bipartite graph $K_{3,3}$ has how many edges?

(a) 6

☒ (b) 9

(c) 12

(d) 3

$3 \times 3 = 9$

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→ colour

24. The chromatic number of a cycle graph C_3 (a triangle) is:

(a) 1

(b) 2

~~(c)~~ 3

(d) 4



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25. A lattice is a POSET in which every pair of elements has:

- (a) Only a LUB
- (b) Only a GLB

- ☒ (c) Both a LUB and a GLB
- (d) No LUB or GLB

LUB = Least upper bound

GLB = Greatest Lower Bound

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26. In Propositional logic, $p \rightarrow q$ is equivalent to:

☒ (a) $\neg p \vee q$

☐ (b) $p \wedge \neg q$

☐ (c) $\neg p \wedge q$

☐ (d) $p \vee q$

$$\neg p \vee q$$
$$F \vee F = F$$

$$P \rightarrow q \equiv \neg p \vee q$$

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27. How many bit strings of length 8 contain exactly three 1s?

(a) 2^8

(b) 8×3

~~(c)~~ $C(8, 3)$

(d) $P(8, 3)$

$${}^8C_3 = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

$${}^nC_r = \frac{n!}{(n-r)!r!}$$

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28. The number of vertices of odd degree in any graph is always:

- (a) Prime
- (b) Odd

-  (c) Even
- (d) Zero

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29. The inverse of the function $f(x) = e^x$ is:

(a) $1/e^x$

(b) $\log(x)$

(c) x

(d) e^{-x}

$$f(x) = e^x, \quad f(x)^{-1} = \log x$$

$$y = e^x$$

$$\log y = \log e^x$$

$$\log y = x$$

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30. Which properties does a 'Ring' satisfy but a 'Group' does not necessarily have?

(a) One binary operation

☒ (b) Two binary operations

(c) Identity element

(d) Closure property